

## Bill of Materials

| Name                                     | Amount | Link                      | Price         |
|------------------------------------------|--------|---------------------------|---------------|
| M4 x 10 mm socket-head cap screws        | 20     | <a href="#">Open link</a> | \$7.99 / pack |
| M3 x 8 mm low-profile button-head screws | 12     | <a href="#">Open link</a> | \$6.99 / pack |
| M3 x 8 mm socket-head cap screws         | 50     | <a href="#">Open link</a> | \$6.99 / pack |
| M3 x 14 mm socket-head cap screws        | 32     | <a href="#">Open link</a> | \$7.99 / pack |
| M3 hex nuts                              | 3      | <a href="#">Open link</a> | \$5.99 / pack |
| M3 x 25 mm socket-head cap screws        | 7      | <a href="#">Open link</a> | \$6.99 / pack |
| M2 x 10 mm socket-head cap screws        | 10     | <a href="#">Open link</a> | \$6.99 / pack |
| M3 x 16 mm socket-head cap screws        | 10     | <a href="#">Open link</a> | \$6.99 / pack |
| M3 x 6 mm socket-head cap screw          | 1      | <a href="#">Open link</a> | \$5.99 / pack |
| M3 x 12 mm socket-head cap screws        | 23     | <a href="#">Open link</a> | \$6.99 / pack |
| M3 x 35 mm socket-head cap screws        | 7      | <a href="#">Open link</a> | \$7.99 / pack |
| M4 x 16 mm socket-head cap screws        | 30     | <a href="#">Open link</a> | \$7.99 / pack |

|                                               |   |                           |                |
|-----------------------------------------------|---|---------------------------|----------------|
| M4 x 14 mm socket-head cap screws             | 5 | <a href="#">Open link</a> | \$7.99 / pack  |
| M4 x 50 mm socket-head cap screws             | 4 | <a href="#">Open link</a> | \$7.99 / pack  |
| M3 x 15/16 mm countersunk screws              | 4 | <a href="#">Open link</a> | \$6.99 / pack  |
| M5 x 16 mm socket-head cap screws             | 6 | <a href="#">Open link</a> | \$7.99 / pack  |
| HTD3M-396 closed-loop timing belt, 6 mm wide  | 1 | <a href="#">Open link</a> | \$9.99         |
| HTD3M-342 closed-loop timing belt, 6 mm wide  | 1 | <a href="#">Open link</a> | \$8.99         |
| HTD3M-201 closed-loop timing belt, 6 mm wide  | 1 | <a href="#">Open link</a> | \$8.99         |
| HTD3M-246 closed-loop timing belt, 6 mm wide  | 1 | <a href="#">Open link</a> | \$8.99         |
| HTD3M 12-tooth pulley, 5 mm bore              | 2 | <a href="#">Open link</a> | \$9.99 / pair  |
| HTD3M 15-tooth pulley, 5 mm bore              | 1 | <a href="#">Open link</a> | \$8.99         |
| 8 mm shaft coupler, M4 mounting holes         | 3 | <a href="#">Open link</a> | \$12.99 / pack |
| AXK3552 thrust needle bearing, 35 x 52 x 4 mm | 1 | <a href="#">Open link</a> | \$10.99        |
| NSK HR32906J tapered                          | 5 | <a href="#">Open link</a> | \$29.99 each   |

|                                             |   |                           |               |
|---------------------------------------------|---|---------------------------|---------------|
| roller bearing                              |   |                           |               |
| NSK HR32907J tapered roller bearing         | 4 | <a href="#">Open link</a> | \$34.99 each  |
| 3 x 8 x 4 mm ball bearing                   | 2 | <a href="#">Open link</a> | \$7.99 / pack |
| 3 x 10 x 4 mm ball bearing                  | 6 | <a href="#">Open link</a> | \$8.99 / pack |
| 12 mm illuminated on/off push button, 3-6 V | 1 | <a href="#">Open link</a> | \$7.99        |
| JST-PH 2.0 mm 4-pin cable/connector         | 1 | <a href="#">Open link</a> | \$7.99 / kit  |
| 24 V cooling fan                            | 1 | <a href="#">Open link</a> | \$8.99        |
| GX16 2-pin male/female power connector      | 1 | <a href="#">Open link</a> | \$8.99 / pair |
| M8 female 4-pin sensor connector            | 1 | <a href="#">Open link</a> | \$9.99        |
| ZW12-3 limit switch                         | 3 | <a href="#">Open link</a> | \$9.99 / pack |
| M4 NPN normally-open inductive sensor       | 1 | <a href="#">Open link</a> | \$13.99       |
| GX-F8A inductive proximity sensor           | 1 | <a href="#">Open link</a> | \$16.99       |
| M5 NPN normally-open inductive sensor       | 1 | <a href="#">Open link</a> | \$13.99       |
| 19 mm emergency-stop button                 | 1 | <a href="#">Open link</a> | \$12.99       |

|                                           |   |                           |                |
|-------------------------------------------|---|---------------------------|----------------|
|                                           |   |                           |                |
| PM4 pneumatic push-in fitting             | 4 | <a href="#">Open link</a> | \$9.99 / pack  |
| 4 x 2.5 mm pneumatic tubing, 2.5 m        | 1 | <a href="#">Open link</a> | \$9.99         |
| XT30 female connector                     | 1 | <a href="#">Open link</a> | \$7.99 / pack  |
| 5/2 pneumatic solenoid valve, 24 V DC     | 1 | <a href="#">Open link</a> | \$18.99        |
| Solenoid valve connector with silencer    | 2 | <a href="#">Open link</a> | \$11.99 / pack |
| PC4-M5 pneumatic fitting                  | 3 | <a href="#">Open link</a> | \$7.99 / pack  |
| 4 mm pneumatic gripper fitting            | 2 | <a href="#">Open link</a> | \$7.99 / pack  |
| Hook-up wire spools                       | 2 | <a href="#">Open link</a> | \$12.99 each   |
| Heat-shrink tubing assortment             | 1 | <a href="#">Open link</a> | \$7.99         |
| Cable gland assortment                    | 1 | <a href="#">Open link</a> | \$9.99         |
| 2-conductor power wire for XT30/GX16      | 1 | <a href="#">Open link</a> | \$12.99        |
| NEMA 17 precision planetary gearbox, 20:1 | 2 | <a href="#">Open link</a> | \$39.99 each   |
| NEMA 17 precision planetary gearbox, 10:1 | 1 | <a href="#">Open link</a> | \$32.99        |
| NEMA 17 stepper                           | 1 | <a href="#">Open link</a> | \$9.99         |

|                                                |   |                           |              |
|------------------------------------------------|---|---------------------------|--------------|
| motor, 16 Ncm, 42 x 42 x 20 mm                 |   |                           |              |
| NEMA 17 stepper motor, 45 Ncm, 42 x 42 x 40 mm | 3 | <a href="#">Open link</a> | \$11.99 each |
| NEMA 17 stepper motor, 65 Ncm, 42 x 42 x 60 mm | 2 | <a href="#">Open link</a> | \$15.99 each |
| 24 V, 5 A power supply                         | 1 | <a href="#">Open link</a> | \$13.99      |

## Final Price

With shipping and taxes, it will come out to \$1,018.99. I am requesting a grant for \$1000 and all other costs will be covered by me.